Manage NVMe services

ONTAP 9.6 REST API reference

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Manage NVMe services

Protocols NVMe services endpoint overview

Overview

A Non-Volatile Memory Express (NVMe) service defines the properties of the NVMe controller target for an SVM. There can be at most one NVMe service for an SVM. An SVM's NVMe service must be created before NVMe host initiators can connect to the SVM.

The Non-Volatile Memory Express (NVMe) service REST API allows you to create, update, delete, and discover NVMe services for SVMs.

Examples

Creating an NVMe service for an SVM

The simpliest way to create an NVMe service is to specify only the SVM, either by name or UUID. By default, the new NVMe service is enabled.

In this example, the `return_records` query parameter is used to retrieve the new NVMe service object in the REST response.

```
# The API:
POST /api/protocols/nvme/services

# The call:
curl -X POST 'https://<mgmt-
ip>/api/protocols/nvme/services?return_records=true' -H 'accept:
application/hal+json' -d '{ "svm": { "name": "svm1" } }'

# The response:
{
"num_records": 1,
"records": [
  {
    "svm": {
      "uuid": "bfb1beb0-dc69-11e8-b29f-005056bb7341",
      "name": "svm1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/bfb1beb0-dc69-11e8-b29f-005056bb7341"
        }
      }
    },
    "enabled": true,
    "_links": {
      "self": {
        "href": "/api/protocols/nvme/services/bfb1beb0-dc69-11e8-b29f-
005056bb7341"
      }
    }
  }
]
}
```

Retrieving the NVMe services for all SVMs in the cluster

```
# The API:
GET /api/protocols/nvme/services

# The call:
curl -X GET 'https://<mgmt-ip>/api/protocols/nvme/services' -H 'accept:
application/hal+json'

# The response:
```

```
{
"records": [
  {
    "svm": {
      "uuid": "ab60c350-dc68-11e8-9711-005056bbe408",
      "name": "svm0",
      "_links": {
        "self": {
          "href": "/api/svm/svms/ab60c350-dc68-11e8-9711-005056bbe408"
        }
      }
    },
    "_links": {
      "self": {
        "href": "/api/protocols/nvme/services/ab60c350-dc68-11e8-9711-
005056bbe408"
      }
    }
  },
  {
    "svm": {
      "uuid": "bfb1beb0-dc69-11e8-b29f-005056bb7341",
      "name": "svm1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/bfb1beb0-dc69-11e8-b29f-005056bb7341"
        }
      }
    },
    "_links": {
      "self": {
        "href": "/api/protocols/nvme/services/bfb1beb0-dc69-11e8-b29f-
005056bb7341"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
    "href": "/api/protocols/nvme/services"
  }
}
}
```

Retrieving details for a specific NVMe service

The NVMe service is identified by the UUID of its SVM.

```
# The API:
GET /api/protocols/nvme/services/{svm.uuid}

# The call:
curl -X GET 'https://<mgmt-ip>/api/protocols/nvme/services/bfb1beb0-dc69-
11e8-b29f-005056bb7341' -H 'accept: application/hal+json'

# The response:
{
"svm": {
  "uuid": "bfb1beb0-dc69-11e8-b29f-005056bb7341",
  "name": "svm1",
  "_links": {
    "self": {
      "href": "/api/svm/svms/bfb1beb0-dc69-11e8-b29f-005056bb7341"
    }
  }
},
"enabled": true,
"_links": {
  "self": {
    "href": "/api/protocols/nvme/services/bfb1beb0-dc69-11e8-b29f-
005056bb7341"
  }
}
}
```

Disabling an NVMe service

Disabling an NVMe service shuts down all active NVMe connections for the SVM and prevents the creation of new NVMe connections.

The NVMe service to update is identified by the UUID of its SVM.

```
# The API:
PATCH /api/protocols/nvme/services/{svm.uuid}

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/protocols/nvme/services/bfb1beb0-
dc69-11e8-b29f-005056bb7341' -H 'accept: application/hal+json' -d '{
"enabled": "false" }'
```

You can retrieve the NVMe service to confirm the change.

```
# The API:
GET /api/protocols/nvme/services/{svm.uuid}

# The call:
curl -X GET 'https://<mgmt-ip>/api/protocols/nvme/services/bfb1beb0-dc69-
11e8-b29f-005056bb7341' -H 'accept: application/hal+json'

# The response:
{
"svm": {
  "uuid": "bfb1beb0-dc69-11e8-b29f-005056bb7341",
  "name": "svm1",
  "_links": {
    "self": {
      "href": "/api/svm/svms/bfb1beb0-dc69-11e8-b29f-005056bb7341"
    }
  }
},
"enabled": false,
"_links": {
  "self": {
    "href": "/api/protocols/nvme/services/bfb1beb0-dc69-11e8-b29f-
005056bb7341"
  }
}
}
```

Deleting an NVMe service

The NVMe service must be disabled before it can be deleted. In addition, all NVMe interfaces, subsystems, and subsystem maps associated with the SVM must first be deleted.

The NVMe service to delete is identified by the UUID of its SVM.

```
# The API:
DELETE /api/protocols/nvme/services/{svm.uuid}

# The call:
curl -X DELETE 'https://<mgmt-ip>/api/protocols/nvme/services/bfb1beb0-
dc69-11e8-b29f-005056bb7341' -H 'accept: application/hal+json'
```

Retrieve NVMe services

GET `/protocols/nvme/services`

Retrieves NVMe services.

Related ONTAP commands

- `vserver nvme show`

Learn more

- DOC /protocols/nvme/services

Parameters

Name	Type	In	Required	Description
svm.uuid	string	query	False	Filter by svm.uuid
svm.name	string	query	False	Filter by svm.name
enabled	boolean	query	False	Filter by enabled
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned.

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records.
records	array[nvme_service]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    }
  }
}
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

_links

Name	Type	Description
self	href	

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

nvme_service

A Non-Volatile Memory Express (NVMe) service defines the properties of the NVMe controller target for an SVM. There can be at most one NVMe service for an SVM. An SVM's NVMe service must be created before NVMe host initiators can connect to the SVM.

An NVMe service is identified by the UUID of its SVM.

Name	Type	Description
_links	_links	

Name	Type	Description
enabled	boolean	The administrative state of the NVMe service. The NVMe service can be disabled to block all NVMe connectivity to the SVM. This is optional in POST and PATCH. The default setting is *true* (enabled) in POST.
svm	svm	SVM, applies only to SVM-scoped objects.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Create an NVMe service

POST `/protocols/nvme/services`

Creates an NVMe service.

Required properties

- `svm.uuid` or `svm.name` - The existing SVM in which to create the NVMe service.

Related ONTAP commands

- `vserver nvme create`

Learn more

- DOC /protocols/nvme/services

Request Body

Name	Type	Description
_links	_links	
enabled	boolean	The administrative state of the NVMe service. The NVMe service can be disabled to block all NVMe connectivity to the SVM. This is optional in POST and PATCH. The default setting is *true* (enabled) in POST.
svm	svm	SVM, applies only to SVM-scoped objects.

Example request

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

Response

```
Status: 201, Created
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records.
records	array[nvme_service]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    }
  }
}
```


Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
1115127	The cluster lacks a valid NVMe license.
2621462	The supplied SVM does not exist.
2621507	NVMe is not allowed for the specified SVM.
2621706	The specified `svm.uuid` and `svm.name` do not refer to the same SVM.
2621707	No SVM was specified. Either `svm.name` or `svm.uuid` must be supplied.
72089650	An NVMe service already exists for the specified SVM.
5374893	The SVM is stopped. The SVM must be running to create an NVMe service.
72089900	An NVMe service cannot be creating in an SVM that is configured for a SAN protocol.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

nvme_service

A Non-Volatile Memory Express (NVMe) service defines the properties of the NVMe controller target for an SVM. There can be at most one NVMe service for an SVM. An SVM's NVMe service must be created before NVMe host initiators can connect to the SVM.

An NVMe service is identified by the UUID of its SVM.

Name	Type	Description
_links	_links	
enabled	boolean	The administrative state of the NVMe service. The NVMe service can be disabled to block all NVMe connectivity to the SVM. This is optional in POST and PATCH. The default setting is *true* (enabled) in POST.
svm	svm	SVM, applies only to SVM-scoped objects.

_links

Name	Type	Description
next	href	
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Delete an NVMe service

DELETE `/protocols/nvme/services/{svm.uuid}`

Deletes an NVMe service. An NVMe service must be disabled before it can be deleted. In addition, all NVMe interfaces, subsystems, and subsystem maps associated with the SVM must first be deleted.

Related ONTAP commands

- `vserver nvme delete`

Learn more

- DOC /protocols/nvme/services

Parameters

Name	Type	In	Required	Description
svm.uuid	string	path	True	

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
2621462	The supplied SVM does not exist.
72089651	The supplied SVM does not have an NVMe service.
72089653	There are subsystems associated with the NVMe service SVM. The subsystems must be removed before deleting the NVMe service.
72089654	There are NVMe-oF LIFs associated with the NVMe service SVM. The LIFs must be removed before deleting the NVMe service.
72090028	The NVMe service is enabled. The NVMe service must be disabled before it can be deleted.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve an NVMe service

GET `/protocols/nvme/services/{svm.uuid}`

Retrieves an NVMe service.

Related ONTAP commands

- `vserver nvme show`

Learn more

- DOC /protocols/nvme/services

Parameters

Name	Type	In	Required	Description
svm.uuid	string	path	True	The unique identifier of the SVM whose NVMe service is to be retrieved.

Name	Type	In	Required	Description
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
enabled	boolean	The administrative state of the NVMe service. The NVMe service can be disabled to block all NVMe connectivity to the SVM. This is optional in POST and PATCH. The default setting is *true* (enabled) in POST.
svm	svm	SVM, applies only to SVM-scoped objects.

Example response

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
2621462	The supplied SVM does not exist.
72089651	The supplied SVM does not have an NVMe service.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update an NVMe service

PATCH `/protocols/nvme/services/{svm.uuid}`

Updates an NVMe service.

Related ONTAP commands

- `vserver nvme modify`

Learn more

- DOC /protocols/nvme/services

Parameters

Name	Type	In	Required	Description
svm.uuid	string	path	True	The unique identifier of the SVM whose NVMe service is to be updated.

Request Body

Name	Type	Description
_links	_links	
enabled	boolean	The administrative state of the NVMe service. The NVMe service can be disabled to block all NVMe connectivity to the SVM. This is optional in POST and PATCH. The default setting is *true* (enabled) in POST.
svm	svm	SVM, applies only to SVM-scoped objects.

Example request

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
1115127	The cluster lacks a valid NVMe license.
2621462	The supplied SVM does not exist.
72089651	The supplied SVM does not have an NVMe service.
5374893	The SVM is stopped. The SVM must be running to create an NVMe service.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

nvme_service

A Non-Volatile Memory Express (NVMe) service defines the properties of the NVMe controller target for an SVM. There can be at most one NVMe service for an SVM. An SVM's NVMe service must be created before NVMe host initiators can connect to the SVM.

An NVMe service is identified by the UUID of its SVM.

Name	Type	Description
_links	_links	
enabled	boolean	The administrative state of the NVMe service. The NVMe service can be disabled to block all NVMe connectivity to the SVM. This is optional in POST and PATCH. The default setting is *true* (enabled) in POST.
svm	svm	SVM, applies only to SVM-scoped objects.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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